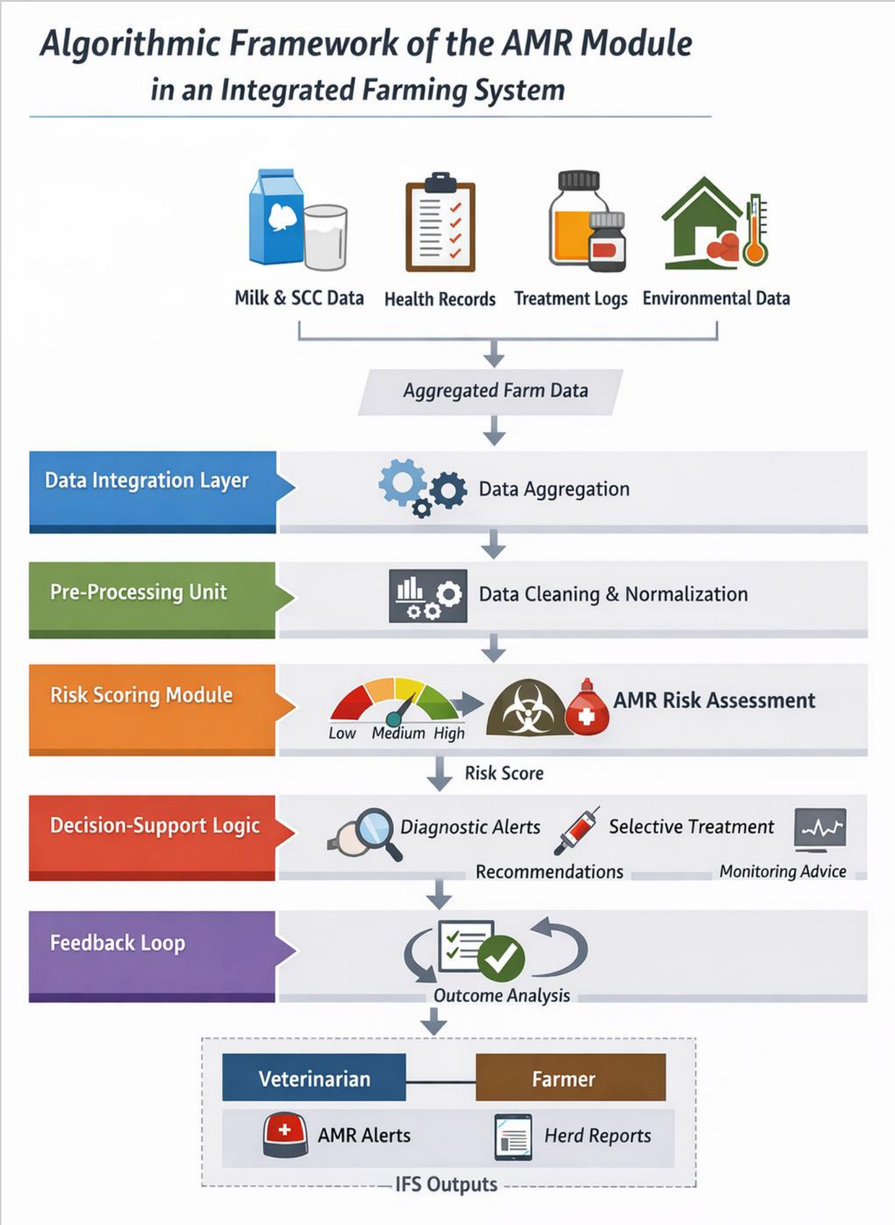
ANTIMICROBIAL RESISTANCE IN MILK PRODUCTION FROM A DATA ANALYTICS PERSPECTIVE: AN ALGORITHMIC MODULE OF AN INTEGRATED FARMING SYSTEM.

Jana Výrostková, Ivana Regecová, Aleš Pavlík, František Zigo, Mária Vargová, Wiktor Bendowski

Univerzita veterinárskeho lekárstva a farmácie v Košiciach

ABSTRAKT

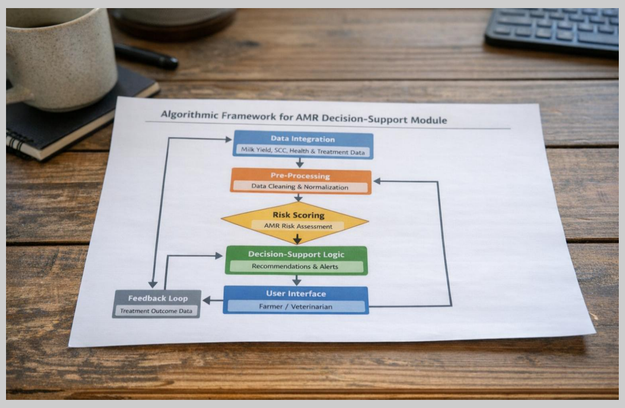
Antimicrobial resistance (AMR) represents a growing challenge in dairy production due to its impact on animal health, milk safety, and public health. Data-driven herd management is increasingly explored as a tool to support responsible antimicrobial use. This pilot-stage study presents a conceptual algorithmic module designed to assist AMR-aware decision-making in dairy herds within an integrated farming system. The framework uses routinely collected herd data, including somatic cell count, health records, and treatment history, to structure a risk-oriented decision-support model. The module aims to identify cows with higher likelihood of repeated antimicrobial exposure and to support selective treatment strategies. Particular emphasis is placed on udder health monitoring, which is closely linked to milk quality and safety. Although field validation is not included at this stage, the study outlines a feasible approach for incorporating AMR considerations into daily herd management using existing farm data. Future research should include on on-farm validation and evaluation of impacts on antimicrobial use and milk quality.



MATERIÁL A METODIKA

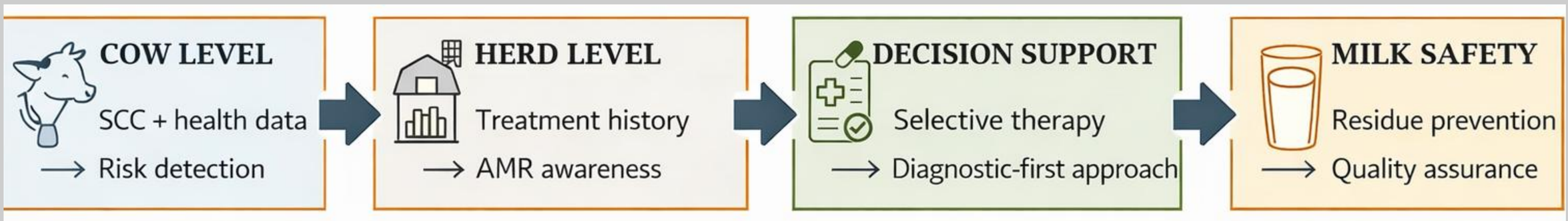
This study was designed as a pilot-stage methodological investigation focusing on the conceptual design and logical structure of an algorithmic module for antimicrobial resistance (AMR) management in dairy herds. The emphasis was placed on framework development rather than on field validation. The objective was to create a scalable model that can be implemented and tested in commercial dairy farm conditions in subsequent research phases.

The proposed module will be evaluated in subsequent studies on selected farms. Validation will include pilot testing under practical farm conditions, comparison of algorithm-generated recommendations with veterinary decision-making, and monitoring of antimicrobial use indicators.



RESULTS

The proposed algorithmic module organizes routinely collected dairy herd data into a structured decision-support framework aimed at reducing unnecessary antimicrobial use while maintaining milk safety. The model integrates somatic cell count (SCC), treatment history, and recorded health events to identify cows with a higher likelihood of repeated antimicrobial exposure. Such an approach enables risk evaluation at both the individual animal and herd levels.



CONCLUSION

This pilot study introduces a conceptual algorithmic module for AMR-aware decision support in dairy herds. The framework demonstrates how routinely collected herd data can be structured to support more selective antimicrobial use while maintaining milk safety.
Further work should focus on on-farm validation and assessment of its practical impact on antimicrobial use and milk quality indicators.

The study was conducted as part of the Visegrad Fund project no. 22430046: Control of antimicrobial resistance by the Integrated Farm System within "Farm to Fork" strategy. The project is co-financed by the governments of Czechia, Hungary, Poland and Slovakia through Visegrad Grants from the International Visegrad Fund. The mission of the fund is to advance ideas for sustainable regional cooperation in Central Europe. This work was also supported by the Slovak Research and Development Agency under the Contract no. APVV-22-0457.